PyGameMaker — Tutorial 10: Turn-Based Multiplayer by File Exchange

How to Use This Handout

This handout goes with the in-app tutorial **Help > Tutorials > Turn-Based Multiplayer: Tic-Tac-Toe by File Exchange** (5 pages, about 25-30 minutes). Keep the tutorial open on one half of your screen; it has the full details of every step. Tick each box when the step is done, and check the **You should see** line at the end of every phase. From Phase 2 you need a partner and a **shared folder** that both computers can use. If something does not match, look at **Stuck?** before you call your teacher.

What You Will Make

Tic-Tac-Toe for two players on two computers, with no live connection: each move is written as a small file in a shared folder, and the other computer reads it. One computer **hosts** (press H) and plays X; the other **joins** (press J) and plays O.

Phase 1: The Board

- [ ] **1.** Create a project TicTacToeFiles with the **Empty Project** template. Create obj_game (no sprite) and a room room_board, and place one obj_game in it.
- [ ] **2.** In obj_game, **Create** event: set nine variables to an empty text: cell_0_0, cell_1_0, cell_2_0, cell_0_1, cell_1_1, cell_2_1, cell_0_2, cell_1_2, cell_2_2 (column first, then row).
- [ ] **3. Draw** event: first *Set draw color* white. Then four *Draw line* actions for the grid: x 270 and 370 from y 90 to 390, and y 190 and 290 from x 170 to 470.
- [ ] **4. Mouse: Left Button Pressed**: nine **If** checks, one per cell, like cell_0_0 == "" and mouse_x >= 170 and mouse_x < 270 and mouse_y >= 90 and mouse_y < 190 then *Set variable* cell_0_0 to X.
- [ ] **5.** In the **Draw** event: for each cell, *If* the variable is not empty, *Draw text* it in the cell (add 100 to x per column and to y per row).

> **You should see:** press **F5**. A white grid appears. Clicking a cell puts an X in it; clicking it again does nothing.

Phase 2: Host a Game / Join a Game

- [ ] **6.** Pick a folder both computers can use (a network drive, or a synced folder). For solo testing, a plain folder name like tictactoe_files is fine.
- [ ] **7. Key press H**: *Host a Game (File Exchange)* with that folder, max players 2, name "Player 1"; then set my_mark to X. **Key press J**: *Join a Game (File Exchange)* with the same folder and name "Player 2"; then set my_mark to O.
- [ ] **8.** In the **Draw** event, before the grid: *If* global.waiting_for_players == 1 then draw "Waiting for opponent..." and *Exit event*; otherwise draw "You are " + my_mark.

> **You should see:** run the game on both computers. Press H on one and J on the other. "Waiting for opponent..." disappears and each screen says "You are X" or "You are O".

Phase 3: Taking Real Turns

- [ ] **9. Create**: set last_round_acted and my_turn to 0.
- [ ] **10. Step**: if it is my turn (global.round_number odd and I am X, or even and I am O) set my_turn to 1, otherwise 0. Then: if my_turn == 0 and last_round_acted != global.round_number, *End Turn (File Exchange)* and set last_round_acted to global.round_number (a "pass").
- [ ] **11.** Rewrite the nine click checks: add my_turn == 1, and instead of setting the variable, *Set a Shared Variable (File Exchange)* with the cell's own name and the value my_mark, then *End Turn (File Exchange)*.
- [ ] **12.** In the **Draw** event, draw each cell from global.cell_0_0 etc. (draw it only if it is not 0; an unset shared variable reads as 0).

> **You should see:** only the player whose turn it is can place a mark. A mark shows on **both** screens a moment after the click, and turns alternate X, O, X.

Phase 4: Winning and Playing Again

- [ ] **13.** In obj_game add the **Round Resolved** event with eight **If** checks, one per line (three rows, three columns, two diagonals): if the three global.cell_* are equal and not 0, set winner to that mark.
- [ ] **14. Create**: set winner to an empty text. **Draw** (last): if winner != "", draw winner + " wins! Press SPACE to play again.".
- [ ] **15. Key press Space**: if winner != "", *Leave the Game (File Exchange)*. Press H and J again for a new game.

> **You should see:** get three in a row and both screens show who won.

Stuck?

What is wrong	Most likely cause	What to do
The screen is empty	The drawing is black on the black room	Add *Set draw color* white at the top of the **Draw** event
It says "Waiting for opponent..." forever	The two computers use different folders, or a computer cannot reach the folder	Use exactly the same folder path; test that both can create a file there
Both computers say "You are X"	Both pressed H	One presses H, the other J
A click does nothing	It is not your turn, the cell is already used, or my_turn is not set	Wait for your turn; check the **Step** event
My mark shows only after a moment	That is normal: it appears after the round resolves (about a second)	Wait about a second
One player's mark never appears	They forgot *End Turn* after the click	Add *End Turn (File Exchange)* to the click handler

Nobody wins even with three in a row	The **Round Resolved** checks are missing a line or use the wrong names	Check all eight lines and the cell names
The names of the cells do not match	A typo, for example cell_1_0 vs cell_0_1	Cell names are column first, then row

Challenges

- **Try this (5 minutes):** change the player names, or the folder.
- **Push further:** reset the nine cells and winner when leaving the game, for a rematch without restarting.
- **Invent:** show a message when the board is full with no winner, or keep a score across several games with another shared variable.

Vocabulary

Term	What it means
Host	The computer that runs the game and decides each round
Shared folder	A folder both computers can read and write
Shared variable	A value that every player can read as global.name
Round	One turn of the game, resolved by the host
Turn-based	Players act one after another, not at the same time

Check Yourself

Write your answers in the notes below.

1. How does a move get from one computer to the other?
2. Why must the player who is not on turn still "pass" with *End Turn*?
3. Why is the mark drawn from global.cell_0_0 and not from the local variable?

My Notes